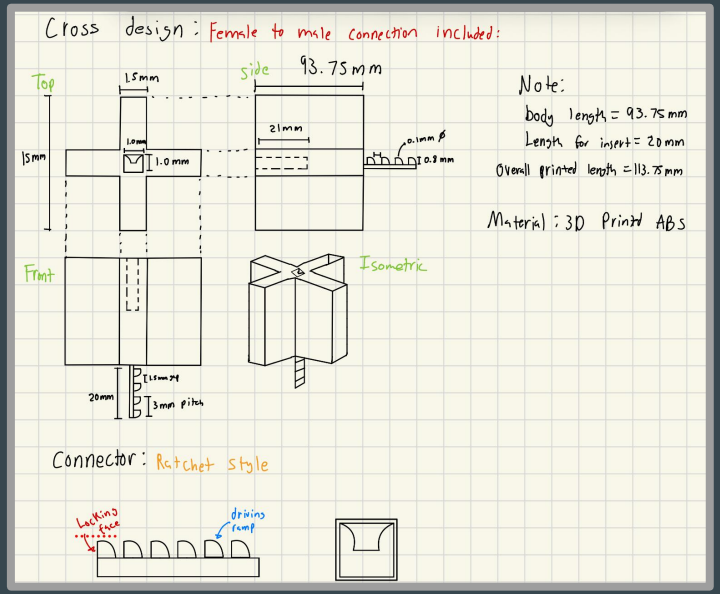
First Design

Pros:

- Cross Beam design
- Ratchet Style connector
- Easy to Assemble
- Simple

Cons:

- ABS Printed
- Thin Ratchet connector prone to snapping while on load
- Bar was too thin

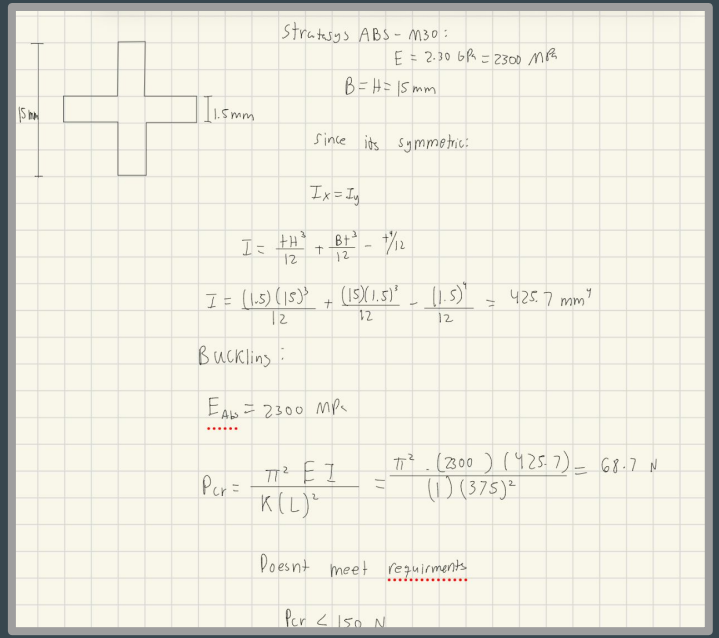


Hand Sketch

Hand Calculations (First Design)

Analysis:

As seen in the hand calculations, the value calculated was less than the required 150 N that assignment called for. The bar had to be redesigned.



Hand Calculations

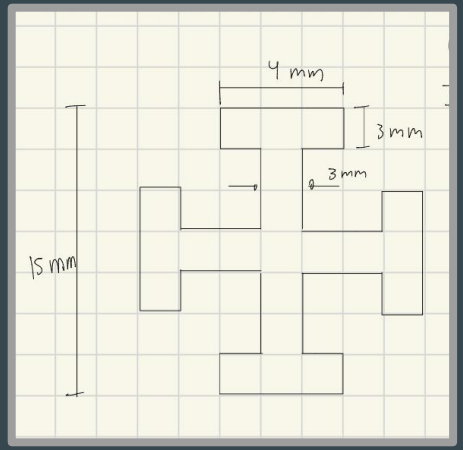
Second Design

Pros:

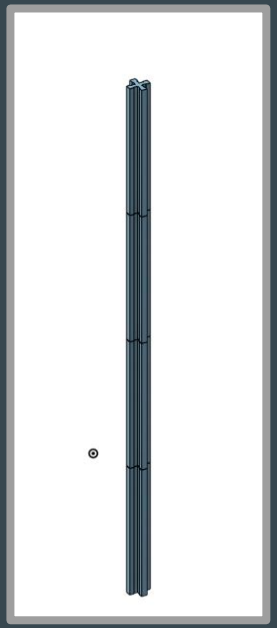
- Stronger against Bending
- Flanges add extra support
- Symmetrical
- Simple slotted pin connection

Cons:

- More Complex
- Use more material
- Sharp Corners more stress concentrators

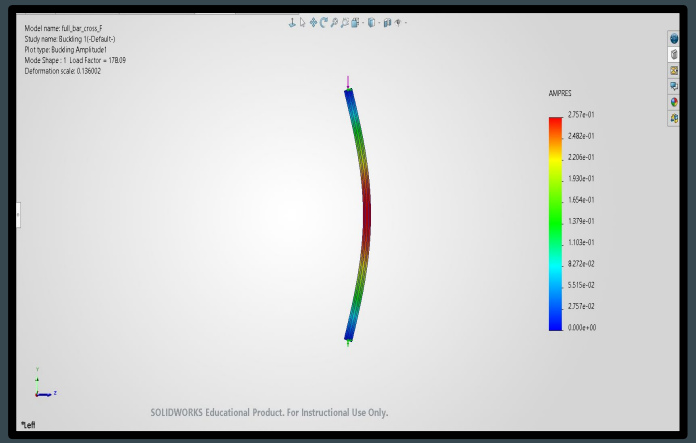


Hand Sketch

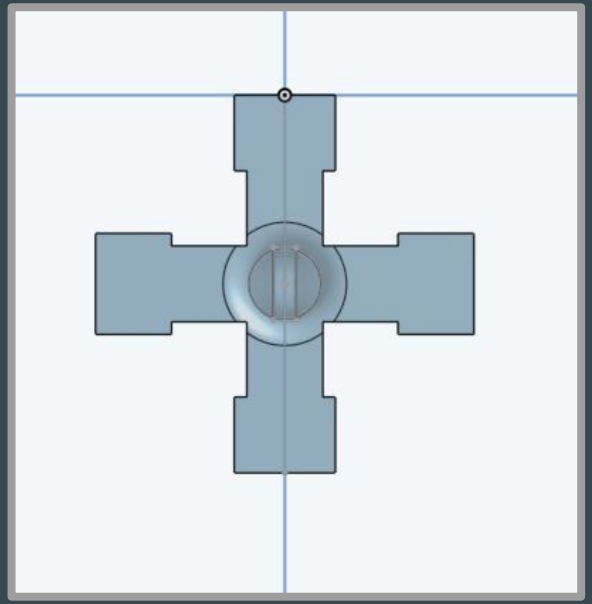


Full bar assembly

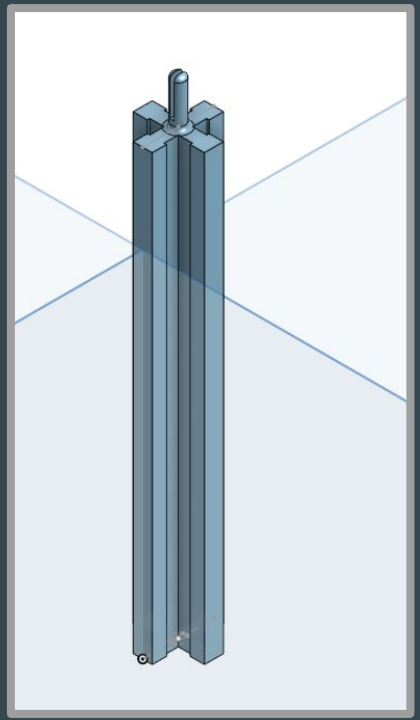
Second Design



Simulation



Female Connection

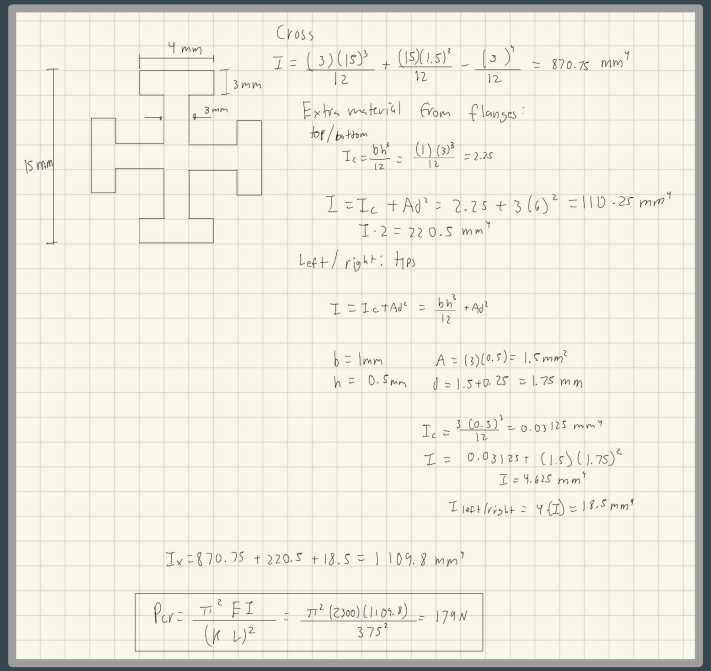


Male Connection

Hand Calculations for Second Design

Analysis:

- N>150 which was P = 179N
- From the simulation P = 178 N
- Mass from Simulation m = 36.6 g
- Hand Calculation for mass m = 36.3 g
- Simple Pin Connection with a slit down the middle to fit together



Hand Calculations

Summary

- First design failed to meet the 150N load
- Bar redesigned with flanges to improve strength
- Second redesign predicted a 179 N buckling
- Solidwork Simulation showed a 178.09 N buckling
- Calculated mass was 36.3 g
- Simulated mass 36.6 g
- Final design successfully exceeded the 150 N load with calculated and simulated values being close to each other.